

DOMESTIC AIRLINES — TRAINING

Hot Weather Operations

Course handbook

Summer and hot weather operations for all fleets: hot-and-high performance, engines and systems, fuel, sand and convective weather, flight safety focus items.

6 lessons · 2 modules · about 0.5 hours

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.

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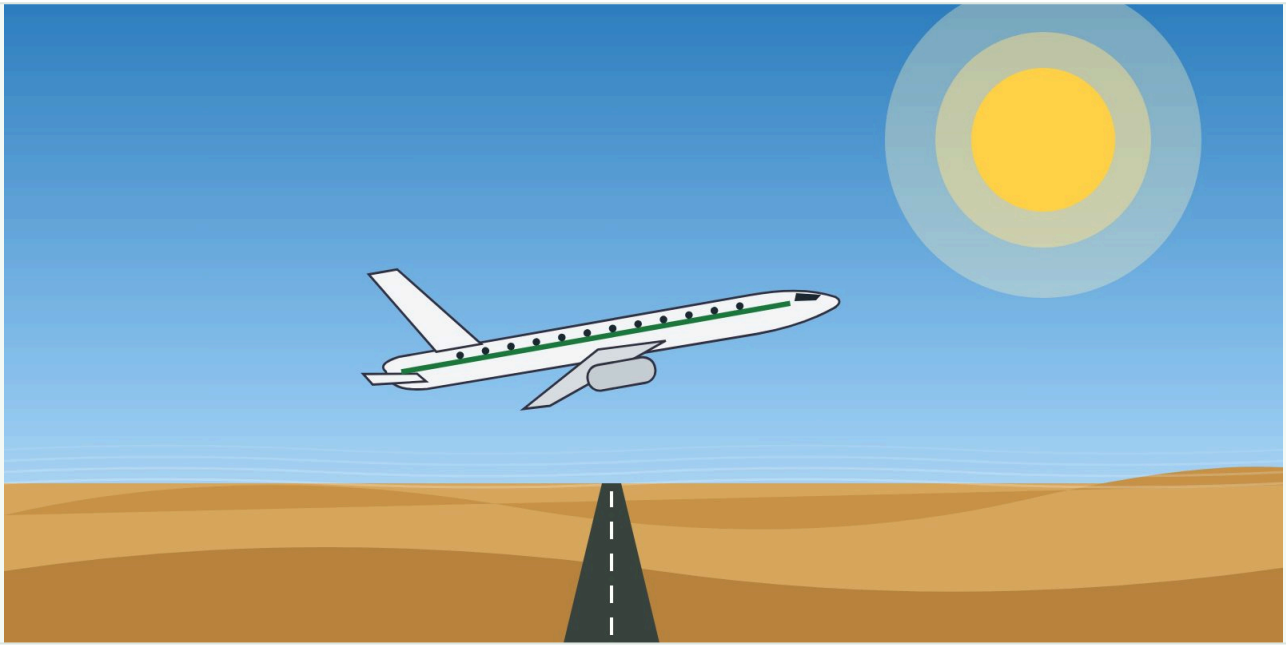
MODULE 1 · PERFORMANCE AND AIRCRAFT SYSTEMS

Lesson 01: Hot and High: Effects on Performance

Lesson 1 of 6 · about 8 min

LEARNING OBJECTIVES

- Explain how temperature and elevation affect aircraft performance.
- Identify the performance limits that become critical in hot weather.
- Use the performance tools correctly.

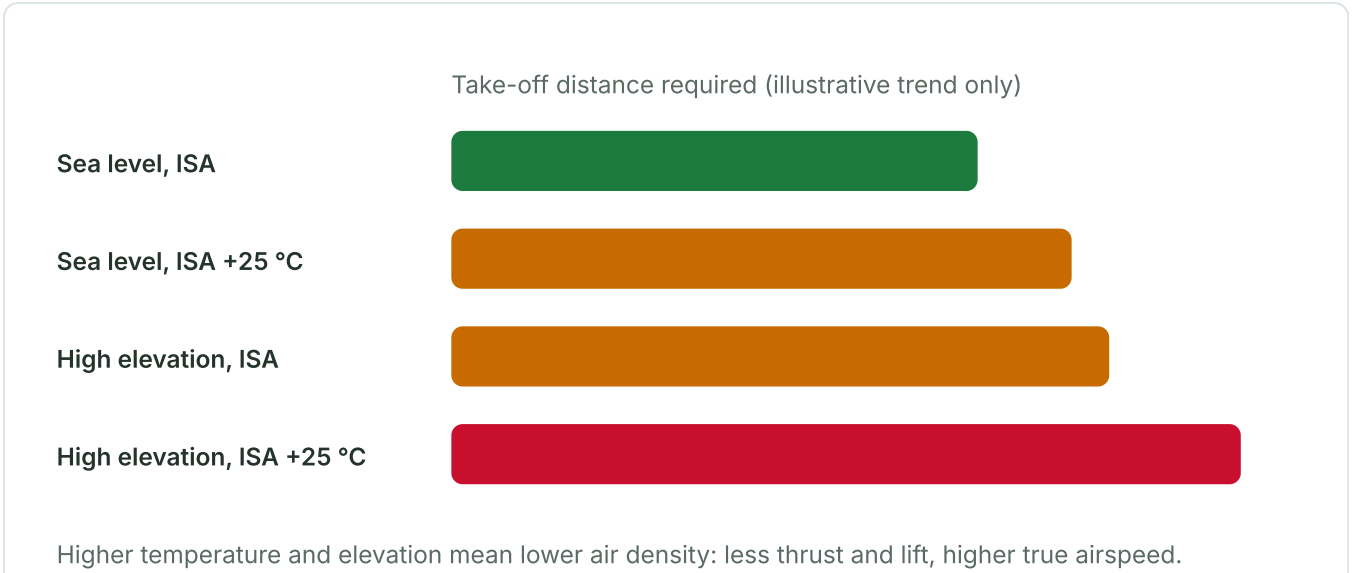


Hot and high: less density, less thrust, longer distances.

Density is everything

High temperature and high aerodrome elevation both reduce **air density** — expressed as a high **density altitude**. Lower density means:

- **less thrust or power** from the engines — jets and turboprops alike are flat-rated only up to a certain temperature;
- **less lift** at a given true airspeed — so the same indicated airspeed means a **higher true airspeed and ground speed**;
- longer take-off and landing distances, higher tyre speeds, and more energy for the brakes to absorb in a rejected take-off;
- reduced **climb gradient** — the climb limit (weight–altitude–temperature) and obstacle clearance often limit the take-off mass on hot days.



Illustrative trend, not performance data. Always use the AFM / performance software, the QRH performance tables or Jetplan for actual values.

Limits that become critical

- **Climb limit** and **obstacle limit** — especially at southern aerodromes with high elevation and temperatures.
- **Runway limit** — check NOTAMs for reduced declared distances (for example, a displaced threshold or the first part of a runway unusable).
- **Tyre speed limit** and **brake energy limit** — higher ground speeds at high density altitude.
- **Landing distance** — higher true airspeed at touchdown; a late or long flare uses runway very quickly.
- **Maximum temperature limits** for take-off and landing in the AFM.

Performance tools

- Use the **approved** performance data: AFM / company performance software, QRH performance tables and the Jetplan operational flight plan.
- Enter the **actual** temperature, QNH, wind and runway condition. A few degrees can change the limiting mass.
- Check aerodrome accessibility: take-off alternate, destination alternate and en-route alternates.
- The first officer prepares the performance dispatch; the captain cross-checks it.

KEY TAKEAWAYS

- Heat and elevation reduce density: less thrust and lift, higher true airspeed and ground speed.
- Climb, obstacle, tyre-speed and brake-energy limits become critical.
- Use approved performance data with actual conditions; check NOTAMs for reduced distances.

KNOWLEDGE CHECK

Why does the same indicated airspeed give a higher ground speed at a hot, high aerodrome?

A. The wind is stronger

B. Lower air density means a higher true airspeed for the same indicated airspeed

C. The engines produce more thrust

D. The runway is longer

Answer: B. At low density, a given indicated airspeed corresponds to a higher true airspeed, and therefore a higher ground speed.

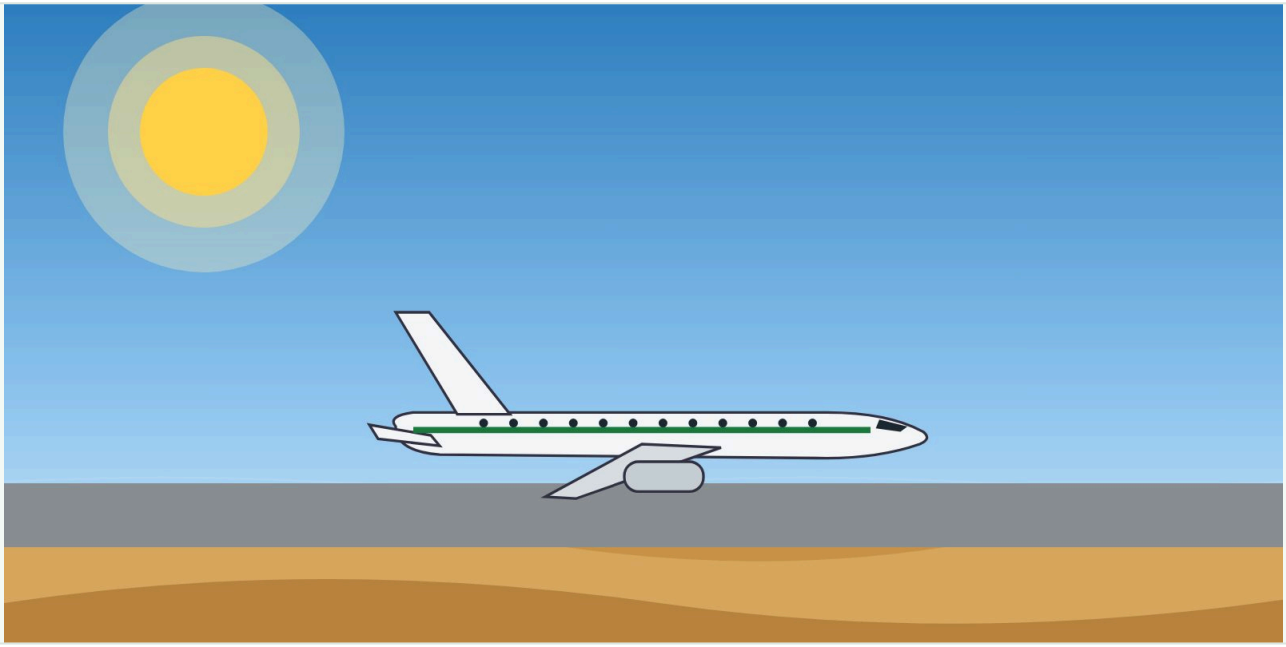
MODULE 1 · PERFORMANCE AND AIRCRAFT SYSTEMS

Lesson 02: Engines, APU and Aircraft Systems in Hot Weather

Lesson 2 of 6 · about 8 min

LEARNING OBJECTIVES

- Apply engine and APU limitations in high temperatures.
- Protect brakes, tyres and avionics in hot conditions.
- Use the fleet-specific hot weather procedures.



Engines, APU, brakes and avionics all have tighter margins in the heat.

Engines and APU

- Exhaust gas / inter-turbine temperature margins are smaller in hot air. Monitor starts closely and know the **start limits** and the hot start and hung start procedures.
- Keep engine running time on the ground to a minimum, and avoid using high thrust on the ground for long periods.
- APU limits (EGT, load) are also closer in hot weather. Ground power and conditioned air reduce APU use.

Brakes and tyres

- Brake temperatures rise faster and cool more slowly in hot weather. Respect the minimum **brake cooling times** between flights.
- Avoid riding the brakes during taxi — keep taxi speeds moderate (Lesson 5).
- Tyre pressures increase with temperature; tyres are also stressed by high ground speeds.

Cockpit, cabin and avionics

- Use ground conditioned air or the APU to cool the cabin and avionics before boarding.
- Watch for avionics overheat cautions and equipment cooling failures.
- Protect crew performance: drink water, manage fatigue and heat stress during long ground times.

Fleet-specific procedures

Topic	Boeing 737-800	Dash 8-Q400 / Q200
Hot weather procedures	FCOM Supplementary Procedures — Adverse Weather, hot weather operation (SP 16.17)	AFM / AOM hot weather and high-temperature limitations
Engine and APU start limits	FCOM limitations; APU start on battery (SP 7.7)	AFM engine start and ITT limits; APU / ground power as installed
Bleeds-off operation	Bleeds-off take-off and landing procedure and performance credit (FCOM / QRH performance)	Bleed and pack settings for take-off per AFM performance charts
Fuel system checks	Spar fuel shut-off valve check (SP 12-01)	AOM fuel system procedures
Performance	QRH performance inflight / dispatch tables; Jetplan	AFM performance charts or company performance software; Jetplan

PREPARE

Before the summer recurrent training, review the hot weather supplementary procedures and limitations for your type. The values and the exact procedure always come from your aircraft manuals.

KEY TAKEAWAYS

- Smaller temperature margins on engine and APU starts — know the limits.
- Respect brake cooling times; avoid riding the brakes.
- Cool the aircraft with ground air or APU; watch avionics cooling.
- Use your type's hot weather procedures (B737 FCOM SP 16.17; Dash 8 AFM/AOM).

KNOWLEDGE CHECK

What is a key consideration for brakes in hot weather?

A. Brakes need no cooling in summer

B. Brake temperatures rise faster and cool more slowly — respect cooling times

C. Use maximum braking on every landing

D. Set the parking brake during taxi

Answer: B. Hot air reduces brake cooling; respect the cooling times between flights.

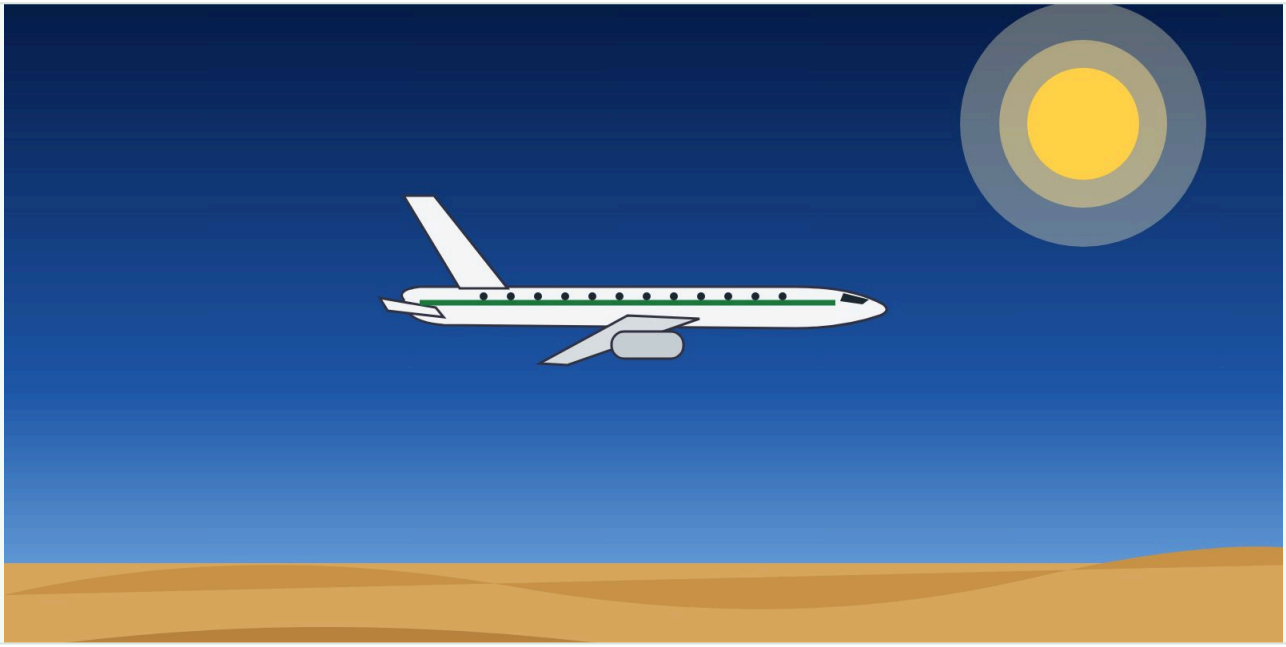
MODULE 1 · PERFORMANCE AND AIRCRAFT SYSTEMS

Lesson 03: Fuel in Hot Weather

Lesson 3 of 6 · about 6 min

LEARNING OBJECTIVES

- Explain the effect of temperature on fuel density and quantity.
- Apply fuel checks and MEL fuel items.
- Use the operational flight plan fuel figures correctly.



Hot fuel is less dense — check the uplift in kilograms.

Fuel density

- Hot fuel is **less dense**: the same mass takes up more volume. Uplift is often measured in litres but planned in kilograms — always convert with the **actual density** on the fuel docket.
- Check the **uplift**: the difference between the calculated and the actual uplift must be within the company tolerance.
- Respect any **maximum fuel temperature** limit in the AFM.

Fuel planning

- Use the Jetplan figures: trip fuel, contingency, alternate, final reserve, taxi and any extra fuel. Check the effect of a change of cruise level or take-off mass.
- Hot weather may impose a lower cruise level or a longer climb. Plan for possible weather deviations around convective activity.
- Check the zero fuel weight and fuel entries carefully — incorrect ZFW or fuel entries are a recurring flight safety event (Lesson 5).

Fuel system and MEL

- B737-800: perform the **spar fuel shut-off valve check** when required (FCOM SP 12-01).
- Read MEL fuel items in full, including the **(M) maintenance** and **(O) operational** procedures.

EXAMPLE FROM THE COMPANY LOFT — B737-800, MEL 28-01-03-01

A main tank aft fuel boost pump may be inoperative provided that: (a) both main tank forward pumps operate normally; (b) at the start of take-off, the fuel quantity in the associated tank is not less than 7,500 lb (3,402 kg); (c) a minimum fuel quantity of 2,500 lb (1,134 kg) is maintained in the associated tank; and (d) the boost pump is deactivated.

Always use the current MEL — this example is for training only.

KEY TAKEAWAYS

- Hot fuel is less dense — convert volume to mass with the actual density.
- Check the uplift and the ZFW / fuel entries carefully.
- Read every MEL item in full, including (M) and (O) procedures.

KNOWLEDGE CHECK

Why must you use the actual fuel density when checking the uplift on a hot day?

A. Fuel is heavier when hot

B. Hot fuel is less dense, so a given volume weighs less

C. Density does not matter

D. To calculate the cabin temperature

Answer: B. Hot fuel occupies more volume for the same mass; the actual density converts litres to kilograms correctly.

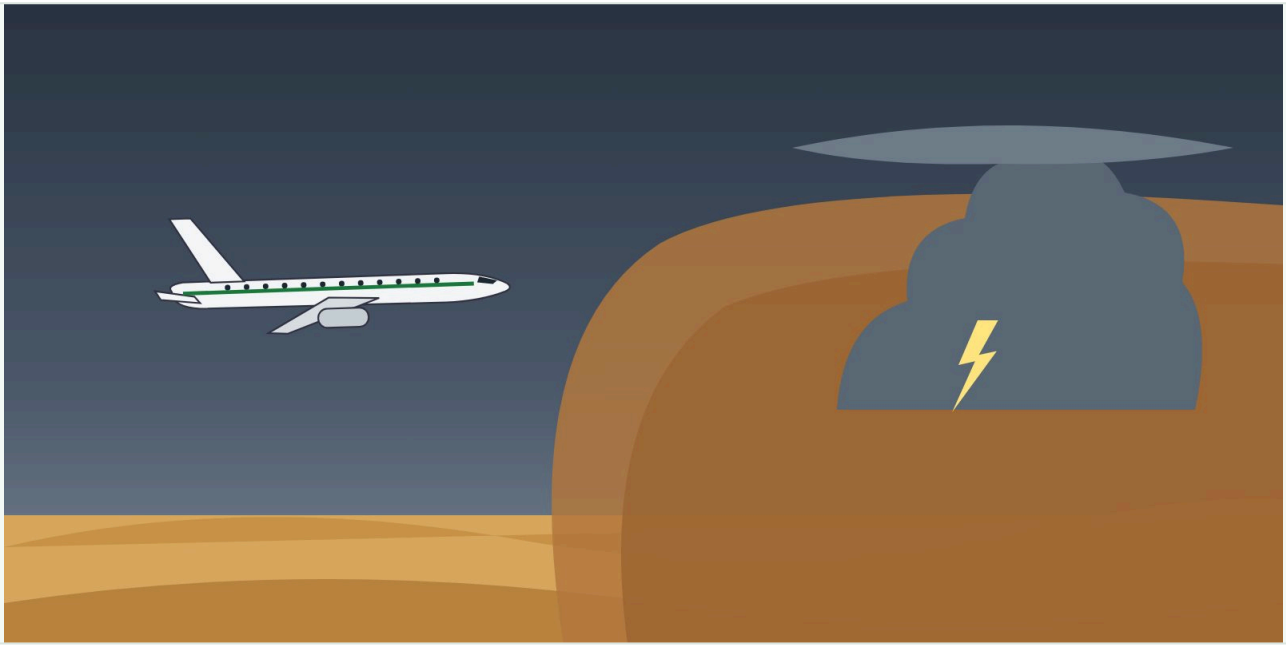
MODULE 2 · WEATHER, SAFETY AND PREPARATION

Lesson 04: Sand, Dust and Convective Weather

Lesson 4 of 6 · about 7 min

LEARNING OBJECTIVES

- Recognise the hazards of dust, sand and drifting sand.
- Apply precautions on the ground and in flight.
- Manage convective weather and gusty winds in summer.



Sand and dust can reduce visibility within minutes.

Dust and sand

- Watch the reports for `DU` (dust), `SA` (sand), `DRSA` (low drifting sand), `BLSA` (blowing sand), `SS` / `DS` (sandstorm / duststorm). Visibility can fall very quickly.
- During the walk-around in dusty conditions, check pitot probes, static ports, engine inlets, APU inlet and air conditioning inlets. Fit covers when the aircraft is parked.
- Minimise engine and APU running and high thrust on sand-covered surfaces to limit ingestion. Use the procedure for taxi and take-off on contaminated or sandy surfaces if published.
- Sand on the runway reduces braking — treat it like a contaminant and report braking action if worse than expected.

Convective weather and wind

- Summer thunderstorms (`CB`, `TCU`) bring gusts, windshear, turbulence and sudden dust storms. Avoid cells by a wide margin.
- **Gusty winds** — for example a reported wind of 18 kt gusting 28 kt — require a careful check of the company crosswind limits and the appropriate speed additives.
- Thermal turbulence close to the ground makes approaches less stable — be ready to go around.

- Dust devils and localised strong winds can affect take-off and landing.

Go-around in hot weather

Climb performance is lower in hot, high conditions. Brief the missed approach, the climb gradient and any obstacles. Know the go-around technique for your type, including engine-out, and practise it in the simulator in hot weather conditions.

KEY TAKEAWAYS

- DU, SA, DRSA, SS/DS: visibility can collapse — watch the trend.
- Protect inlets and probes; minimise ground running and high thrust on sand.
- Summer CB and gusts: wide avoidance, check crosswind limits and speed additives.
- Lower go-around performance in hot, high conditions — brief it.

KNOWLEDGE CHECK

What does DRSA in a METAR mean?

A. Dry runway, sand removed

B. Low drifting sand

C. Drizzle and sand

D. Dust storm

Answer: B. DRSA = low drifting sand.

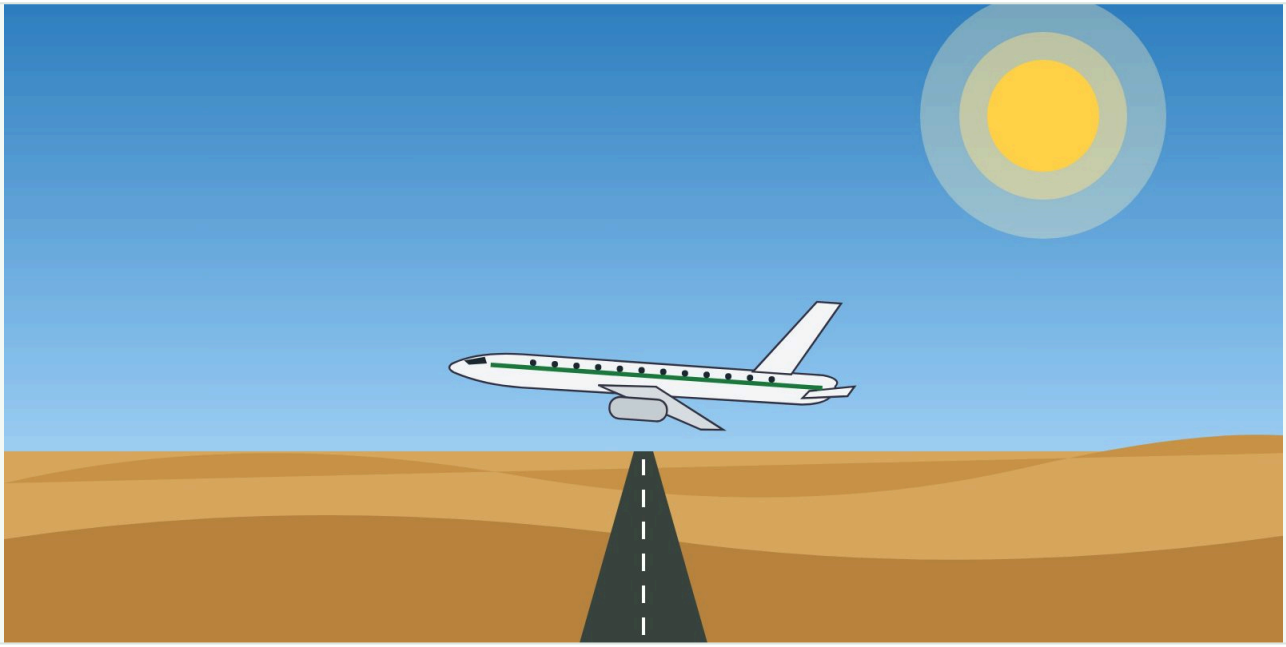
MODULE 2 · WEATHER, SAFETY AND PREPARATION

Lesson 05: Summer Flight Safety Focus Items

Lesson 5 of 6 · about 7 min

LEARNING OBJECTIVES

- Review the flight safety events selected for this season.
- Apply the correct technique to prevent each event.
- Report events through the safety reporting system.



Stabilised approach, normal flare — go around if a long landing develops.

The training department has selected the following flight safety items for discussion with the TRI/TRE:

Event	Why it matters	Prevention
High taxi speed	Hot brakes, reduced margins on turns and runway crossings, passenger injury.	Respect the company taxi speeds; use idle thrust and brief, firm braking applications rather than riding the brakes.
Late or long flare	Each second of float at high ground speed uses a large part of the runway and increases the landing distance.	Fly a stabilised approach to the aiming point; flare at the normal height; go around if a long landing develops.
Incorrect flap setting, ZFW or fuel entries	Wrong take-off speeds and trim — tail strike or degraded take-off performance.	Independent cross-check of performance data, FMS / performance entries and the load sheet by both pilots.
Hard landing	Structural damage, injuries; often after an unstable approach or late flare.	Stabilised approach; correct flare; report any suspected hard landing in the technical log.
Bounced landing	Risk of tail strike, nose gear damage or a second hard landing.	Apply the company bounced landing recovery technique; go around if in doubt.

Use of reverse and APU	Covered by the company reminder note (note de rappel).	Apply the reminder note exactly as published.

Safety notes and reporting

- Read and discuss the latest company **safety notes** before the simulator session.
- **Report events** through the safety reporting system. Reports are used to improve training and procedures, under a just culture.

KEY TAKEAWAYS

- Moderate taxi speeds protect brakes and people.
- A late or long flare at high ground speed eats runway — go around if needed.
- Cross-check flap, ZFW and fuel entries.
- Apply the bounced and hard landing techniques; report every event.

KNOWLEDGE CHECK

What is the most effective prevention for a long landing developing after a floated flare?

A. Push the nose down to land

B. Go around

C. Increase reverse thrust only

D. Extend the flaps further

Answer: B. If a long landing develops, the safe action is a go-around.

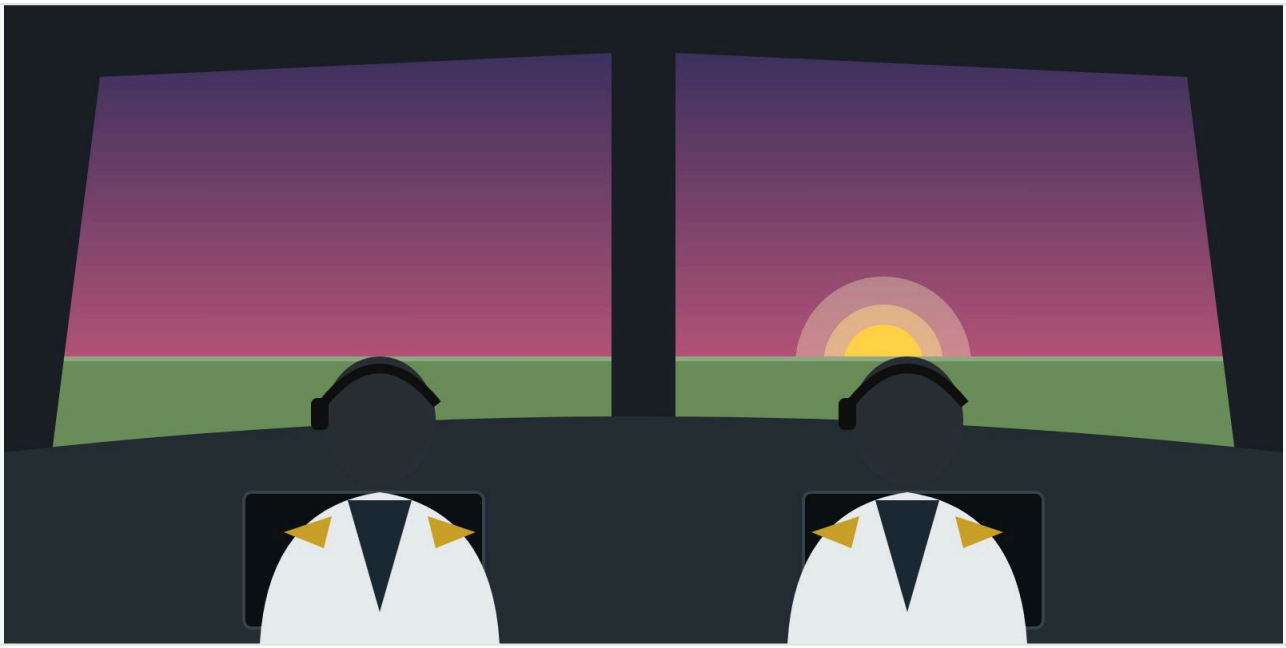
MODULE 2 · WEATHER, SAFETY AND PREPARATION

Lesson 06: MEL, Dispatch and the Summer Simulator Session

Lesson 6 of 6 · about 6 min

LEARNING OBJECTIVES

- Apply the MEL and dispatch criteria for a summer flight.
- Prepare the summer recurrent simulator session.
- Combine hot weather, performance and LOFT topics.



Prepare the summer recurrent simulator session as a crew.

MEL and dispatch

- Check every MEL item for its effect on performance — some items impose mass penalties or limitations that matter more in hot weather.
- Apply the (M) and (O) procedures, the placards and any dispatch conditions.
- The MEL also contains the dispatch requirements for systems that need a specific approval — RVSM, RNAV/PBN and LVO.

The summer recurrent programme

- **Day 1 — ground refresher**: engine and APU operation and limitations in hot weather, hot weather procedures, fuel, APU start on battery, bleeds-off take-off and landing, QRH performance, fuel system checks, RVSM, MEL and dispatch.
- **Days 2 and 3 — simulator**: LOFT sectors to and from hot, high southern aerodromes with dust and drifting sand, MEL dispatch items, go-arounds and engine failures in hot weather, an RNAV approach briefing including temperature effects, and an LVO approach.

- **Mandatory exercises**: upset recovery, windshear identification and escape, TCAS, EGPWS and pilot incapacitation — performed according to company procedures and DACM recommendations.

Preparing

- Prepare the performance dispatch from the Jetplan in the briefing room and discuss it as a crew.
- Review the courses in this academy: CFIT and EGPWS, RVSM, Upset Prevention and Recovery.

KEY TAKEAWAYS

- Check MEL items for performance penalties — critical in hot weather.
- Summer recurrent: ground refresher day, then two simulator days with LOFT and mandatory exercises.
- Prepare performance and review the related academy courses before the session.

KNOWLEDGE CHECK

Which exercises are mandatory in the summer simulator session?

A. Only engine failure

B. Upset recovery, windshear, TCAS, EGPWS and pilot incapacitation

C. Only CAT II approaches

D. None

Answer: B. Upset recovery, windshear, TCAS, EGPWS and pilot incapacitation are mandatory.

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